In [1]:

```
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from tkinter import Tk, filedialog
import ipywidgets as widgets
from IPython.display import display

%matplotlib widget

def process_file(file_path):
    try:
        print(f"Loading data from {file_path}...")
        data = pd.read_excel(file_path, sheet_name="Sheet4")

        required_columns = ["L1", "V2", "AVF"]
        for col in required_columns:
            if col not in data.columns:
                raise KeyError(f"Required column '{col}' not found in the dataset."

        x = data["V2"].values
        y = data["L1"].values
        z = -data["AVF"].values

        if len(x) == 0 or len(y) == 0 or len(z) == 0:
            raise ValueError("One or more columns are empty.")

        num_points = len(x)

        if num_points % 3 == 0:
            color_categories = np.repeat([0, 1, 2], num_points // 3)
        else:
            color_categories = np.repeat([0, 1, 2], num_points // 3 + 1)[:num_point

        fig = plt.figure(figsize=(8, 6))
        ax = fig.add_subplot(111, projection='3d')

        colors = ['green', 'orange', 'red']
        labels = ['Early', 'Mid', 'Late']

        for category in range(3):
            indices = np.where(color_categories == category)[0]
            ax.scatter(
                x[indices], y[indices], z[indices],
                color=colors[category],
                label=labels[category],
                marker='o', s=50
            )

        ax.set_xlabel('V2')
        ax.set_ylabel('L1')
        ax.set_zlabel('-AVF')
        ax.set_title(os.path.basename(file_path))
        ax.legend()
        ax.view_init(elev=30, azim=60)

        out = widgets.Output()
        with out:
            plt.show()
        display(out)

    except FileNotFoundError:
        print(f"File not found: {file_path}. Please check the file path and try aga
    except KeyError as e:
        print(f"Column name '{e.args[0]}' not found in the dataset.")
    except ValueError as e:
        print(f"Error: {str(e)}")
    except Exception as e:
        print(f"An unexpected error occurred: {str(e)}")
```

```
def process_folder(folder_path):
    excel_files = [
        f for f in os.listdir(folder_path)
        if f.endswith('.xlsx') or f.endswith('.xls')
    ]

    if not excel_files:
        print("No Excel files found in the selected folder.")
        return

    print(f"Found {len(excel_files)} Excel file(s). Processing...")

    for file_name in excel_files:
        file_path = os.path.join(folder_path, file_name)
        process_file(file_path)


def main():
    root = Tk()
    root.withdraw()
    root.attributes('-topmost', True)  # Brings dialog to front
    folder_path = filedialog.askdirectory(title="Select Folder")
    root.destroy()

    if not folder_path:
        print("No folder selected. Exiting...")
        return

    process_folder(folder_path)


if __name__ == "__main__":
    main()
```

```
Found 1 Excel file(s). Processing...
Loading data from C:/Users/hp/Desktop/New folder (2)\new_Abdul Barik 20 Control.xls
x...
Output()
```

In [ ]: